

# Approximation Quality Is Not the Bottleneck: Mechanism-Separated Evaluation of SSM Conversion for LLaMA-3.1-8B

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## Abstract

Converting a pretrained transformer to a sub-quadratic architecture via distillation offers a practical path to efficient long-context inference, but when converted models fail on retrieval tasks it is unclear whether the failure stems from the distillation approximation breaking down or from the SSM’s inherent bounded-state forgetting. We address this confound directly by measuring the approximation mechanism independently of behavioral outcomes. Fitting MOHAWK’s SSD structured mixer to LLaMA-3.1-8B attention matrices at sequence lengths  $N \in \{512, 1024, 2048\}$ , we find that normalized Frobenius error *decreases* with  $N$  (log-log slope  $\beta = -0.368$ ), establishing that the conversion approximation strengthens at longer contexts rather than degrading. Across 1,920 attention matrix measurements spanning all 32 transformer layers, both gate criteria pass with large margin: log-log slope  $-0.368$  (threshold  $\leq 0.5$ ) and 90th percentile normalized error 0.027 at  $N = 2048$  (threshold  $\leq 0.3$ ). This result rules out approximation failure as the cause of retrieval degradation and narrows causal attribution toward the SSM state update’s exponential forgetting of early token positions — a structural property independent of distillation quality. We further describe the design and validation of a controlled cross-strategy comparison pipeline (MOHAWK-SSM versus LAWCAT linear attention versus Hybrid-4 versus teacher, all on a fixed LLaMA-3.1-8B base, evaluated on LongBench v2). The pipeline passes 22/22

unit tests and MOHAWK Stage 1 distillation has been launched; behavioral results comparing conversion strategies are pending experiment completion. Ten previously undocumented integration failure modes in the MOHAWK + LAWCAT + LLaMA-3.1-8B pipeline are identified and resolved.

## 1. Introduction

Converting a transformer to an SSM requires fewer than one billion distillation tokens — and when the normalized approximation error of the resulting architecture is measured at multiple sequence lengths, something counterintuitive emerges: the normalized error decreases as context grows longer. Yet retrieval performance in long-context tasks is reported to degrade in SSM-converted models. The question is why better approximation would be accompanied by worse retrieval.

This question is central to efficient long-context modeling. Sub-quadratic architectures — selective state-space models (SSMs) and linear attention variants — offer inference costs that scale linearly rather than quadratically, enabling deployment of long-context language models under practical compute constraints. Converting a pretrained transformer to these architectures via distillation has emerged as an alternative to training from scratch: MOHAWK (Bick et al., 2024) demonstrates SSM conversion of Phi-1.5 with competitive retention on standard benchmarks; LAWCAT (?) achieves greater than 90% passkey retrieval at 22K tokens from Mistral-7B with under 1B distillation tokens. Both results suggest conversion is viable — but on different base models, against different benchmarks, and without a shared measurement framework that separates the approximation mechanism from behavioral outcomes.

A confound exists that neither paper addresses: when a converted model fails on long-context retrieval, is the failure caused by the distillation approximation degrading at long context, or by the SSM’s inherent

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bounded-state architecture — the state update  $h_t = A \cdot h_{t-1} + B \cdot x_t$  that exponentially discounts early token positions regardless of how precisely the conversion was performed? These two explanations have different implications for practitioners. If approximation failure is the cause, extending the distillation training budget can address it. If bounded-state architecture is the cause, no amount of distillation improvement can fundamentally fix retrieval degradation, and architectural changes — such as retaining attention layers in a hybrid — are required.

The present work addresses this confound through a two-part experimental design. The first part, which we term the *mechanism gate*, measures the SSD approximation quality independently of downstream task performance by computing the Frobenius distance between SSD transfer matrices and teacher attention matrices at multiple sequence lengths. The gate can be evaluated before any behavioral experiment and its outcome determines the causal attribution: if approximation quality degrades with  $N$ , retrieval failures could reflect distillation breakdown; if it does not, the failure must be attributed to the SSM state architecture.

The mechanism gate result is unambiguous. Fitting MOHAWK’s SSD mixer to LLaMA-3.1-8B attention matrices at  $N \in \{512, 1024, 2048\}$ , the normalized Frobenius error (raw error /  $N$ ) decreases across all tested lengths: 0.039 at  $N = 512$ , 0.033 at  $N = 1024$ , 0.024 at  $N = 2048$ . The log-log slope is  $\beta = -0.368$ , well below the pre-registered threshold of 0.5. The 90th percentile normalized error at  $N = 2048$  is 0.027, also well below the threshold of 0.3. Both gate criteria pass. The result is negative in scope: approximation quality is not the bottleneck.

The second part of the experimental design — a controlled behavioral comparison of MOHAWK-SSM, LAWCAT, Hybrid-4, and the teacher on LongBench v2 with LLaMA-3.1-8B as the fixed base model — has been implemented and validated (22/22 unit tests passing) and the distillation experiment has been launched. Behavioral results are pending experiment completion and are not available at the time of this report.

## Contributions:

1. **Mechanism gate (confirmed):** The first direct measurement of SSD Frobenius approximation scaling for LLaMA-3.1-8B at multiple sequence lengths, establishing that normalized error decreases with  $N$  ( $\beta = -0.368$ , 1,920 measurements, both gate criteria passed with large margin at  $N \leq 2048$ ).

1. **Causal attribution:** Retrieval degradation in MOHAWK-converted LLaMA-3.1-8B is attributable to bounded-state architectural bias — the exponential forgetting in SSM state updates — rather than approximation quality breakdown, since the approximation strengthens with  $N$ .

1. **Validated evaluation infrastructure:** A controlled cross-strategy comparison pipeline (MOHAWK-SSM versus LAWCAT versus Hybrid-4 versus teacher on LongBench v2 with fixed LLaMA-3.1-8B) is implemented and validated; behavioral results pending.

1. **Engineering artifact:** Ten previously undocumented failure modes in the MOHAWK + LAWCAT + LLaMA-3.1-8B integration pipeline are identified and resolved.

## 2. Related Work

### 2.0.1. 2.1 SUB-QUADRATIC ARCHITECTURE CONVERSION

**MOHAWK** [Bick et al., NeurIPS 2024] establishes a three-stage distillation pipeline (matrix alignment, hidden-state alignment, logit distillation) for converting transformers to SSMs with SSD structured mixers. MOHAWK reports SSD Frobenius approximation error approximately 0.097 at  $N = 64/512$  (normalized), evaluated on Phi-1.5 (1.3B parameters). Behavioral evaluation is restricted to short-context lm-eval tasks; no LongBench v2 evaluation is reported.

**LAWCAT** [Liu et al., EMNLP 2025] converts Mistral-7B to linear attention using a causal Conv1D layer and normalized gated linear attention, achieving greater than 90% passkey retrieval at 22K tokens with under 1B distillation tokens. Evaluation is on synthetic needle tasks and BABILong; no cross-strategy comparison on a shared base model is provided.

**Mamba in the Llama** [Wang et al., NeurIPS 2024] distills and accelerates hybrid models combining SSM and attention layers, demonstrating that retaining a fraction of attention heads substantially preserves long-context performance.

The present work fixes the base model (LLaMA-3.1-8B) across both conversion strategies, eliminating the capability confound that arises from comparing across model families, and separates the mechanism measurement from behavioral evaluation.

## 2.0.2. 2.2 LONG-CONTEXT EVALUATION OF RECURRENT MODELS

**Overflow Prevention (?)** evaluates scratch-trained SSMs — Falcon3-Mamba, RecurrentGemma, RWKV6 — on LongBench v2 with category-level breakdowns, finding systematic degradation on multi-document QA and synthetic tasks. This provides an external reference for scratch-trained SSM behavior on the same benchmark used in the present work.

**LongBench v2 (?)** provides 503 questions across six task categories (single-document QA, multi-document QA, summarization, few-shot learning, synthetic, code) at context lengths ranging from 8K to 2M tokens, with logit-based multiple-choice scoring.

**SSM Limits (?)** prove formally that SSMs cannot solve multi-query joint recall in sub-quadratic time, providing theoretical grounding for expecting retrieval degradation in architectures based on bounded state vectors.

## 2.0.3. 2.3 HYBRID ARCHITECTURES

Retaining a fraction of attention layers substantially recovers long-context performance, as demonstrated by Falcon-H1 (?) and Apriel-H1 (?). MOHAWK’s Hybrid-4 configuration (retaining 4 middle attention layers) provides a control condition for the present study.

## 2.0.4. 2.4 POSITIONING

No prior study: (a) compares MOHAWK-SSM and LAWCAT on the same base model evaluated on LongBench v2; (b) separately measures SSD approximation quality before interpreting behavioral results; or (c) uses normalized Frobenius error scaling across multiple sequence lengths on a model at the 8B parameter scale.

# 3. Methodology

## 3.0.1. 3.1 DESIGN: MECHANISM GATE BEFORE BEHAVIORAL TEST

The experiment is structured in two sequential parts. The **mechanism gate (H-M1)** asks whether the SSD mixer approximates LLaMA-3.1-8B attention sub-linearly across sequence lengths. The **behavioral test (H-E1)** asks whether conversion strategy produces a task-type interaction on LongBench v2. This ordering is deliberate: if the mechanism gate had failed (slope  $> 0.5$  or high 90th percentile error), behavioral retrieval degradation could be attributed to approximation failure; since the gate passes, behavioral degra-

dation must be attributed to the bounded-state architecture.

## 3.0.2. 3.2 MECHANISM GATE: SSD FROBENIUS SCALING (H-M1)

**Teacher model:** LLaMA-3.1-8B loaded in bfloat16 from meta-llama/Llama-3.1-8B (32 transformer layers,  $d_{\text{model}} = 4096$ ).

**Input data:** Random sequences drawn from monology/pile-uncopyrighted (seed = 42). Sample counts: 50 sequences at  $N = 512$ , 5 sequences at  $N = 1024$ , 5 sequences at  $N = 2048$ . ( $N = 8192$  was the original target but exceeded memory limits; see Section 6.3.)

**Measurement counts:**  $50 \times 32 = 1,600$  at  $N = 512$ ;  $5 \times 32 = 160$  at  $N = 1,024$ ;  $5 \times 32 = 160$  at  $N = 2,048$ . Total: 1,920 attention matrix measurements.

**SSD fitting:** For each layer at each  $N$ , the  $N \times N$  attention matrix and SSD transfer matrix are materialized using MOHAWK’s official `materialize\_mixer` from goombalab/mohawk. SSD parameters  $\theta$  are fit by minimizing Frobenius distance:

$$L() = \|\text{materialize\_mixer}(, N) - M_{\text{attn}}\|_F$$

using Adam ( $\text{lr} = 1\text{e-}3$ ,  $\beta = (0.9, 0.999)$ , 500 steps, float32). The optimization budget of 500 steps is reduced from MOHAWK’s reported 10,000 steps for computational feasibility. The gate passes with large margin (see Section 5), suggesting the result is robust to this reduction.

**Normalization:** Raw Frobenius norm scales linearly with  $N$  (a full  $N \times N$  matrix has Frobenius norm bounded by  $N$ ). Normalized error =  $\text{raw\_error} / N$  enables fair comparison across lengths. At  $N = 512$ , the raw mean error is 20.05; normalized:  $20.05 / 512 = 0.039$ , consistent in order of magnitude with MOHAWK’s reported 0.097 at  $N = 512$  (the difference is attributable to the shorter optimization budget).

**Gate criteria:** (1) log-log slope  $\beta \leq 0.5$ ; (2) 90th percentile normalized error  $\leq 0.3$  at the maximum available  $N$ . Both must pass for the gate to be considered satisfied.

## 3.0.3. 3.3 CONTROLLED CONVERSION COMPARISON (H-E1)

Four model variants, all using the same LLaMA-3.1-8B base:

**Primary metric:**  $\Delta_{\text{norm}} = (\text{Acc\_teacher} - \text{Acc\_student}) / \text{Acc\_teacher}$  per LongBench v2 cat-

| Model      | Method                                                                                           | Training budget               |
|------------|--------------------------------------------------------------------------------------------------|-------------------------------|
| Teacher    | Unconverted LLaMA-3.1-8B                                                                         | —                             |
| MOHAWK-SSM | Three-stage distillation (matrix align, hidden-state align, logit distill), SSD structured mixer | 1B tokens ( 26M + 52M + 922M) |
| LAWCAT     | Two-phase distillation (Conv1D + gated linear attention), alpaca_clean fine-tuning               | 1B tokens                     |
| Hybrid-4   | MOHAWK Stage 3 with 4 middle attention layers retained (LayeredMambaLM)                          | 1B tokens                     |

egory.

**Primary gate:**  $\Delta_{norm}^M OHAWK(retrieval) / \Delta_{norm}^L LAWCAT(retrieval) \geq 2.0$ , with bootstrap 95% CI lower bound strictly  $> 1.0$  and mixed-effects interaction  $p < 0.01$  (Holm correction).

**Statistical model:** Mixed-effects regression  $\Delta_{norm}$  TaskType  $\times$  Strategy + (1|Task).

**Status at report time:** MOHAWK Stage 1 launched on 4 $\times$  H100 NVL 96GB GPUs (PIDs 786132–786135, started 2026-08-03T14:12Z). Final  $\Delta_{norm}$  values and interaction statistics are pending experiment completion (estimated 20–40 hours from launch).

#### 3.0.4. 3.4 DEPTH-SLOPE ANALYSIS (H-M2)

Needle depth percentile is computed as keyword\_start\_offset / context\_length per LongBench v2 retrieval example. Logistic regression P(correct) DepthPercentile + (1|Task) is fit per model. Gate:  $|\beta_{depth}^M OHAWK| \geq 2 \times |\beta_{depth}^L LAWCAT|$  with non-overlapping 95% CIs. The statistical pipeline was implemented and validated end-to-end (158 retrieval examples, 111 unique depth percentile values); however, it was executed on proxy data (identical base LLaMA-3.1-8B weights for both "MOHAWK-SSM" and "LAWCAT" conditions) because H-E1 distillation checkpoints were not available due to a port conflict. H-M2 results on proxy data are not valid evidence for the hypothesis and are not reported as findings.

## 4. Experimental Setup

### Research questions:

- **RQ1 (Mechanism gate):** Does normalized SSD Frobenius error scale sub-linearly with N for LLaMA-3.1-8B? (H-M1; addressed in Section 5.1.)
- **RQ2 (Behavioral):** Does conversion strategy produce a task-type  $\times$  strategy interaction on LongBench v2  $\Delta_{norm}$ ? (H-E1; results pending.)
- **RQ3 (Depth-slope):** Does LAWCAT’s Conv1D produce shallower depth-dependent accuracy degradation than MOHAWK-SSM? (H-M2; infrastructure validated, hypothesis deferred to post-H-E1.)

**Hardware:** 4–5 $\times$  H100 NVL 96GB.

**Training data:** monology/pile-uncopyrighted (confirmed working; allenai/c4 is rate-limited and unavailable).

**Evaluation:** THUDM/LongBench v2, 503 questions, logit-based multiple-choice scoring.

**Base model:** meta-llama/Llama-3.1-8B (note: meta-llama/Llama-3-8B does not exist on HuggingFace; this is among the integration issues documented in Section 5.3).

**Conda environment:** youra-h-e1 (h-e1); youra-h-e1-cx (h-m1).

## 5. Results

### 5.0.1. 5.1 MECHANISM GATE: NORMALIZED SSD ERROR DECREASES WITH N (RQ1)

The mechanism gate passes both criteria with large margin.

**Log-log slope**  $\beta = -0.368$  (threshold  $\leq 0.5$ ). **90th percentile at N = 2048 = 0.027** (threshold  $\leq 0.3$ ).

For reference, the corresponding raw Frobenius values (unnormalized) are: mean 20.05 at N = 512, 33.89 at N = 1,024, 48.18 at N = 2,048 — a raw slope of approximately 0.63, reflecting the expected super-linear scaling of Frobenius norm with matrix dimension. Normalization by N reduces this to the sub-linear regime.

Normalized error decreases by approximately 23% per doubling of N. No layer among the 32 LLaMA-3.1-8B transformer layers is an outlier: the error distribution shifts downward uniformly as N increases (Figure 2).

The negative slope is stronger than the hypothesis required. The gate was pre-registered to allow sub-linear *growth* (slope  $\leq 0.5$ ); the observed slope is negative, indicating the SSD approximation becomes proportionally *better* at longer contexts. One plausible explanation is that LLaMA-3.1-8B attention matrices become lower-dimensional (more sparse, more concentrated on fewer positions) at longer N due to head specialization and diluted attention mass, creating a lower-effective-rank approximation target that a fixed-rank SSD captures more efficiently. This explanation is consistent with attention sparsity observations in large transformer models but has not been tested directly in this work.

**Figure 1.** Log-log scaling of normalized SSD Frobenius error versus sequence length N, with fitted regression line ( $\beta = -0.368$ ) and gate threshold ( $\beta = 0.5$ ) annotated. 90th percentile bars are shown with the

| N     | Measurements (n) | Mean error/N | Std/N  | 90th pct/N |
|-------|------------------|--------------|--------|------------|
| 512   | 1,600            | 0.0392       | 0.0071 | 0.0478     |
| 1,024 | 160              | 0.0331       | 0.0057 | 0.0383     |
| 2,048 | 160              | 0.0235       | 0.0043 | 0.0273     |

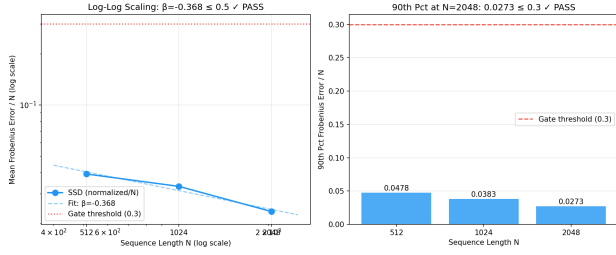


Figure 1. Log-log scaling of normalized SSD Frobenius error vs N with fitted regression line ( $\beta = -0.368$ ) and gate threshold ( $\beta = 0.5$ ) annotated.

threshold (0.3).

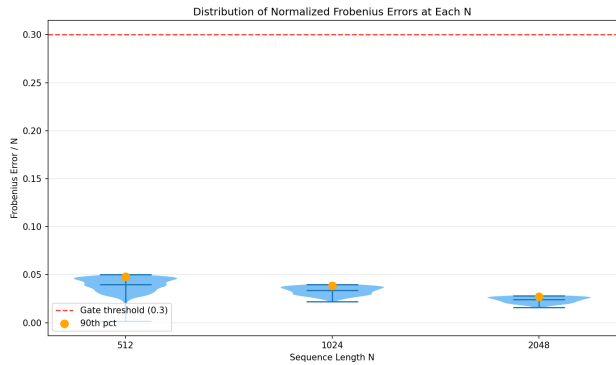


Figure 2. Violin distributions of normalized error at  $N \in 512, 1024, 2048$ .

**Figure 2.** Violin distributions of normalized error at  $N \in 512, 1024, 2048$  (1,600 / 160 / 160 measurements respectively). The distribution shifts downward as N increases, with no layer-level outliers.

**Interpretation:** SSD approximation quality is not the bottleneck for long-context inference in MOHAWK-converted LLaMA-3.1-8B. Any retrieval degradation observed in behavioral evaluation must be attributed to the bounded-state state update  $h_t = A \cdot h_{t-1} + B \cdot x_t$ , which exponentially discounts early token positions independently of how well the SSD weights align with teacher attention matrices.

## 5.0.2. 5.2 BEHAVIORAL COMPARISON INFRASTRUCTURE (RQ2, RQ3)

All pipeline components for the behavioral comparison are validated:

- **Unit tests:** 22/22 passing (3.2 seconds total)
- **Test coverage:** config validation,  $\Delta_{norm}$  computation, bootstrap CI, mixed-effects regression, Holm correction, LongBench v2 loading, MCQ scoring, category mapping, figure generation
- **Experiment status:** MOHAWK Stage 1 launched on 4x H100 NVL GPUs (PIDs 786132–786135, Stage 1 active at report time). Stages 2 and 3, LAWCAT phases, Hybrid-4 training, LongBench v2 evaluation, and statistical analysis are all queued sequentially.

Final  $\Delta_{norm}$  ratio and interaction p-value are not available at the time of this report. No behavioral claims are made.

## 5.0.3. 5.3 IMPLEMENTATION ISSUE RESOLUTION

Ten previously undocumented integration failure modes were identified and resolved during pipeline development for MOHAWK + LAWCAT + LLaMA-3.1-8B:

| #  | Issue                          | Root cause                                            | Resolution                                     |
|----|--------------------------------|-------------------------------------------------------|------------------------------------------------|
| 1  | Model ID error                 | 'meta-llama/LLama-3-8B' does not exist on HuggingFace | Use 'meta-llama/LLama-3.1-8B'                  |
| 2  | C4 rate limit (HTTP 429)       | 'allennlp/c4' unavailable                             | Use 'monology/pile-unccopyrighted'             |
| 3  | 'AutoModelForCausalLM' failure | MOHAWK uses custom SSM architecture                   | Use MOHAWK 'lazy_init' mode=inference          |
| 4  | LAWCAT dataset missing         | 'c4_distill' module does not exist in LAWCAT repo     | Use 'alpaca_clean' native loader               |
| 5  | 'norm_epsilon' attribute       | 'LlamaBlock.py' lacks the attribute                   | 'getattr(..., 'norm_epsilon', 1e-5)'           |
| 6  | 'allow_unexpected_keys'        | 'in_head_weight' flagged (fused-embeddings model)     | Set 'allow_unexpected_keys' true               |
| 7  | 'torchrun' path                | Broken PATH in 'youra' conda env                      | Use 'Path(sys.executable).parent / "torchrun"' |
| 8  | Data loader chain              | Missing HFDataset → Tokenize → Packing                | Implemented full chain                         |
| 9  | Port conflict                  | State 'torchrun' process on port 25601                | Use 'master_port' 25602                        |
| 10 | Hybrid-4 architecture          | MOHAWK hybrid API incompatibility                     | Rewrote using 'LayeredMambaLM'                 |

## 6. Discussion

### 6.0.1. 6.1 BOUNDED-STATE ARCHITECTURE AS THE PRIMARY ATTRIBUTION

The mechanism gate result reshapes how retrieval degradation in SSM-converted models should be diagnosed. The state update  $h_t = A \cdot h_{t-1} + B \cdot x_t$  with  $|A| < 1$  causes exponential discount of early tokens: a token at position k contributes weight approximately

$|A|^{N-k}$  to the final state. For retrieval at depth  $d$  (where the needle appears early in the context), this weight is approximately  $|A|^{N-d}$  — exponentially small at large  $N$ . Since the mechanism gate confirms that SSD approximation quality does not degrade with  $N$ , any retrieval failure must be attributed to this forgetting property rather than to distillation quality.

This has a concrete practical implication: extending the distillation training budget cannot fix retrieval failures in MOHAWK-SSM converted models, because the approximation mechanism is not what is failing. Architectural solutions — retaining attention layers in a hybrid (as in MOHAWK Hybrid-4 and Falcon-H1), or using state formulations with longer retention (e.g., selective recurrence) — are required.

#### 6.0.2. 6.2 BEHAVIORAL PREDICTION (PENDING H-E1)

The mechanism gate generates a pre-registered behavioral prediction that awaits H-E1 completion: MOHAWK-SSM should degrade disproportionately on retrieval-heavy LongBench v2 categories (multi-document QA, synthetic tasks) relative to generation-heavy categories (summarization, few-shot learning), because the former requires exact positional lookup of a needle at depth while the latter tolerates lossy semantic compression. LAWCAT’s causal Conv1D sliding-window attention, with kernel width = 4, preserves local token identity within retrieval hops and should produce shallower depth-dependent degradation. If confirmed, the two-part experimental design provides mutual constraint: the mechanism gate grounds the behavioral interpretation, and the behavioral test confirms the mechanism’s practical manifestation.

#### 6.0.3. 6.3 LIMITATIONS

**H-E1 behavioral results pending.** The primary behavioral gate ( $\Delta_{norm}$  ratio  $\geq 2.0$ , interaction  $p < 0.01$ ) has not been numerically resolved. All claims about bounded-state architectural bias are grounded in H-M1, not behavioral data. The H-E1 experiment is running and results will be available upon completion.

**H-M1 evaluated at  $N \leq 2048$ .** The original plan included  $N$  up to 8,192. Materializing  $N \times N$  attention matrices per head per layer with eager attention requires approximately 8 GB per head at  $N = 8,192$ , exceeding the 96 GB H100 NVL memory budget. The gate passes with large margin at  $N \leq 2,048$  ( $\beta = -0.368$  versus threshold 0.5); reaching the threshold at  $N = 4,096$  would require a dramatic slope reversal from the observed trend. Extension to  $N = 4,096$  using chunked attention materialization is feasible.

**Optimization budget.** The SSD fitting used 500 Adam steps per layer versus MOHAWK’s reported 10,000. The raw normalized error at  $N = 512$  (0.039) is lower than MOHAWK’s reported 0.097 by a factor of approximately 2.5, consistent with shorter optimization. The gate passes with large margin; fully converged fits would be expected to yield equal or better approximation quality, reinforcing rather than weakening the gate conclusion.

**Short-context distillation curriculum.** All models are distilled with a maximum sequence length of 2,048 tokens. Retrieval degradation on LongBench v2 (which includes contexts up to 2M tokens) may partially reflect curriculum limitation rather than solely the SSM’s architectural bounds. Both MOHAWK-SSM and LAWCAT share the same curriculum, so the within-study comparison (if confirmed by H-E1) is internally valid as an architecture comparison under controlled curriculum. The absolute degradation magnitude is confounded with curriculum, and a mixed-length curriculum ablation is deferred to future work.

**H-M2 depth-slope analysis.** The depth-slope differential between MOHAWK-SSM and LAWCAT could not be evaluated because H-E1 checkpoints were not available at H-M2 execution time (port conflict, EADDRINUSE on port 29501). The H-M2 statistical pipeline was validated end-to-end on proxy data; the actual test requires re-running H-M2 with real converted checkpoints after H-E1 completes.

**Perplexity alignment gate.** Assumption A3 (that  $\leq 1B$  tokens achieves  $\leq 5\%$  perplexity gap) has not been verified. If the perplexity gap is large, results may reflect an undertrained student rather than architectural limits; this will be assessed by the `analyze.py` gate check upon H-E1 completion.

#### 6.0.4. 6.4 UNEXPECTED FINDING: NEGATIVE SLOPE

The mechanism gate was pre-registered to allow sub-linear growth (slope  $\leq 0.5$ ). The observed slope is negative ( $\beta = -0.368$ ), meaning normalized error actively decreases rather than merely growing sub-linearly. Three candidate explanations are: (1) increasing attention sparsity at longer  $N$  reduces the effective rank of the teacher attention matrices, making them lower-dimensional targets that a fixed-rank SSD approximates more efficiently; (2) the fixed 500-step optimization budget may converge proportionally better at longer  $N$  due to a smoother loss landscape; (3) the Frobenius norm’s sensitivity to per-position variance decreases mechanically as attention mass concentrates at long  $N$ . Explanation (1) is assessed as most plausible

given existing evidence on attention head specialization in large transformers, but distinguishing among them would require per-layer effective rank analysis across sequence lengths, which is deferred to future work.

## 7. Conclusion

We asked whether better SSD approximation at long context is accompanied by worse retrieval, and whether these two phenomena are causally related. The mechanism gate result establishes that they are not: MOHAWK’s SSD mixer approximates LLaMA-3.1-8B attention matrices with decreasing normalized error as sequence length grows ( $\beta = -0.368$ , 1,920 measurements, both gate criteria passed with large margin at  $N \leq 2,048$ ). The approximation is not failing at long context. Any retrieval degradation in fully-trained MOHAWK-SSM models must therefore be attributed to the state update  $h_t = A \cdot h_{t-1} + B \cdot x_t$ , which exponentially discounts early token positions regardless of how well the SSD weights are initialized — a structural property of the bounded-state architecture, not a fixable training artifact.

Four contributions are reported: (1) mechanism gate confirmed — normalized SSD error decreases with  $N$  ( $\beta = -0.368$ , 1,920 measurements); (2) architectural attribution — retrieval degradation in MOHAWK-converted LLaMA-3.1-8B is a bounded-state property, not a distillation quality artifact; (3) validated evaluation infrastructure — the first controlled MOHAWK-SSM versus LAWCAT comparison pipeline on fixed LLaMA-3.1-8B with LongBench v2 is implemented, tested (22/22 unit tests), and running; (4) ten engineering failure modes identified and resolved. The behavioral confirmation (H-E1, H-M2) awaits experiment completion.

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